

## ML LAB EXPT – 2

### Aim of the Experiment

Develop a program to compute the correlation matrix to understand the relationships between pairs of features. Visualize the correlation matrix using a heatmap to know which variables have strong positive/negative correlations. Create a pair plot to visualize pairwise relationships between features. Use California Housing dataset.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

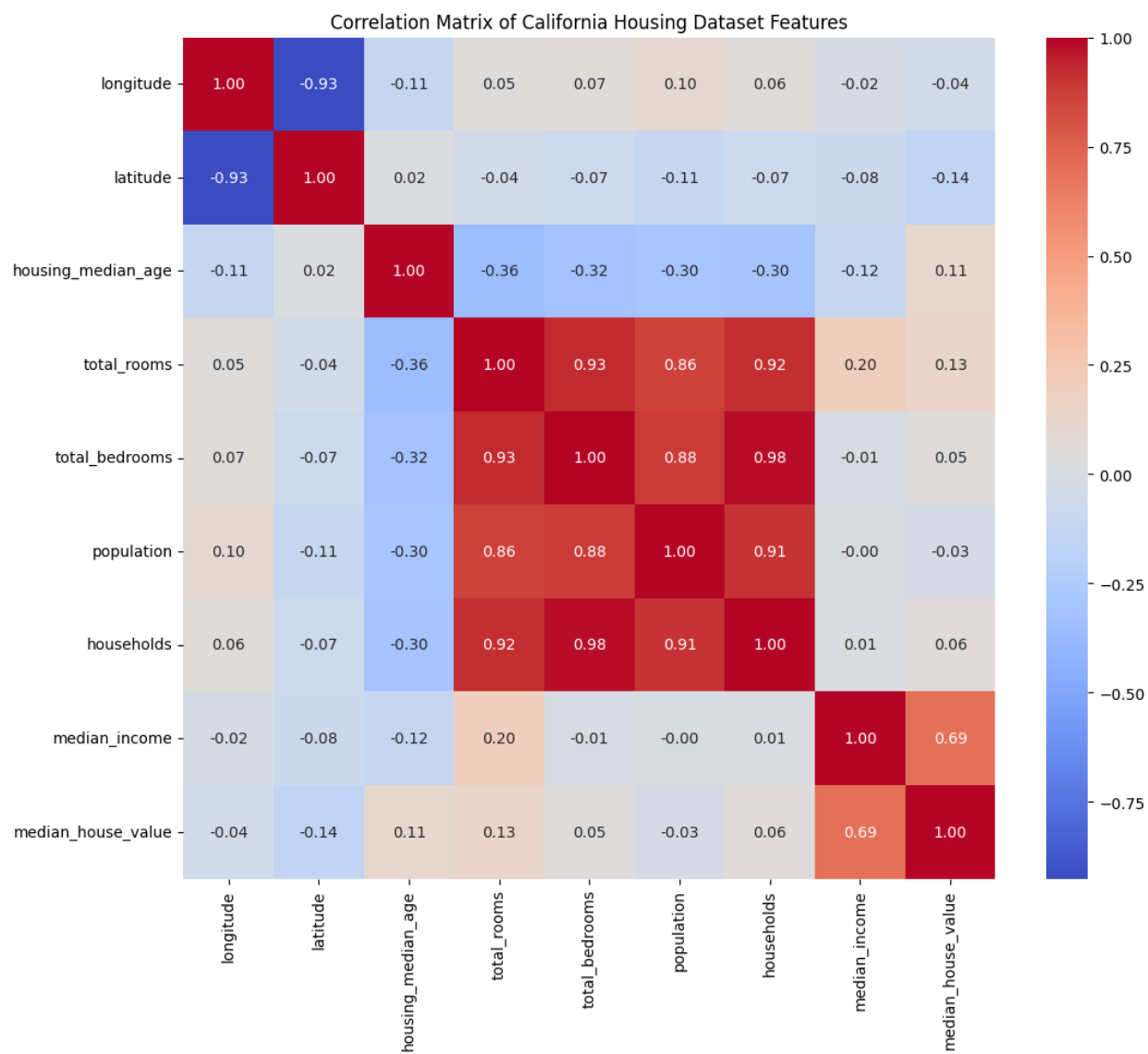
# Load the California Housing dataset
try:
    df = pd.read_csv('california_housing_train.csv')
except FileNotFoundError:
    print("Error: california_housing_train.csv not found")
    exit()

# Compute the correlation matrix
correlation_matrix = df.corr()

# Visualize the correlation matrix using a heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix of California Housing Dataset Features')
plt.show()

# Create a pair plot to visualize pairwise relationships
sns.pairplot(df)
plt.suptitle('Pairwise Relationships between Features', y=1.02)
plt.show()
```

Output:



Pairwise Relationships between Features

